What is a Multicore Processor?

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- A processor that has more than one core is called Multicore Processor. In this article we will learn about multicore processor in detail. A multicore processor is an integrated circuit for faster processing of several tasks like decreased power consumption and increased performance. it has two or more processors that read and perform program instructions.

In other words, a multicore processor comprises multiple processing units, or "Cores," each of which has the potential to do distinct tasks. For example, if you are doing many tasks at a time such as watching a movie and using other applications. one core processor will handle activities like watching a movie while the multicore processor handles other responsibilities at the same time.

History of Multicore Processors

The companies who created the chip based processors could only put one CPU on one chip. Chipmakers were able to create chip with more circuits as chip making technology also chipmakers were able to generate multi core chip with more than one processor. The first multicore CPU was invented by Kunle Olukotun in 1998 who was a professor of electrical engineering at Stanford. Multicore chips were first accessible in 2005 from Advanced Micro Devices and Intel.

Uses of Multicore Processor

Multicore processors are used in many devices like desktop, laptop, smartphone, and gaming systems. Some applications which use multicore processor are as below.

- Multicore Processor is used in high graphics games like Overwatch and Star Wars Battlefront, and other 3D games.
- The multicore processor is more appropriate used in video editing software like Adobe Photoshop, and iMovie.
- Multicore Processor is used solidworks with computer-aided design (CAD).
- Database servers also handled by multicore CPU.
- Multicore CPU is used in high network traffic.
- Embedded systems can handle by multicore processor.

Architecture of Multicore Processor

A multi-core processor's design enables communication between all existing cores, and they divide and assign all processing duties appropriately. each core's processed data is through back to the Motherboard by a single common gateway once all of the operations have been finished. In terms of total performance this technique beats a single core CPU.

Advantages of Multicore Processor

- **Performance:** A multi-core CPU can perform more work as compared to a one-core processor. so multi core processor performance is better.
- **Reliability:** The software is always assigned to different cores in multi core processor. If one piece of software fails others remain unaffected.
- **Software Interactions:** If a software is running on many cores. it will communicate with each other.
- **Multitasking:** Multi core CPU can perform multiple tasks at a same time even if many applications may be run at the same time.

- **Power Consumption:** A multi-core processor consumes less power. Only a part of the CPU that produces heat will be used. Due to low battery utilization the power consumption is automatically reduced.

Disadvantages of Multicore Processors

- **Application Speed:** A multi-core processor is designed for multitasking, its performance is not enough. when an software is processing It jump from one core to the next. so the result is the cache fills up, increasing its speed.
- **Jitter:** In a multi-core CPU, the cores increase more interference happens due to this result in excessive jitters. operating system's program performance may suffer, and failures frequency may increase .
- **Analysis:** When you are doing multiple task at once, you will need to add memory. In a multi-core CPU, this is tough to analysis it.
- **Resource Sharing:** A multi-core CPU shares multiple resources, both internal and external. Like networks, system buses, and main memory are some resources. Any software running on the same core again and again it will interrupted.

Conclusion

In summary, multicore processors enable previously unprecedented levels of performance, scalability, and efficiency. They constitute a major forward in computer technology. Multicore processors have revolutionized the way we approach complicated computational tasks by utilizing the power of parallelism and sharing computational resources among multiple cores. This has opened up new opportunities in different fields, from consumer electronics to scientific research.

Difference between MultiCore and MultiProcessor System

In today's tech world, multi-core and multi-processor systems have become essential for boosting computing power and efficiency. A multicore system packs several processing units or cores into a single chip allowing it to tackle multiple tasks at once. Multiprocessor system uses two or more separate processors that share resources enabling them to work together seamlessly. This means that if one processor runs into trouble the others can keep things running smoothly. Both systems are designed to meet the increasing demands of modern applications with multicore systems focusing on maximizing performance within a single chip and multiprocessor systems expanding capabilities through multiple CPUs.

What is a Multicore System?

A processor that has more than one core is called a Multicore Processor while one with a single core is called Unicore Processor or Uniprocessor. Nowadays, most systems have four cores (Quad-core) or eight cores (Octa-core). These cores can individually read and execute program instructions, giving feel like a computer system has several processors but in reality, they are cores and not processors. Instructions can be calculation, data transferring instruction, branch instruction, etc. Processors can run instructions on separate cores at the same time. This increases the overall speed of program execution in the system. Thus heat generated by the processor gets reduced and increases the overall speed of execution.

Multicore systems support MultiThreading and Parallel Computing. Multicore processors are widely used across many application domains, including general-purpose, embedded, network, digital signal processing (DSP), and graphics (GPU). Efficient software algorithms should be used for the implementation of cores to achieve higher performance. Software that can run parallel is preferred because we want to achieve parallel execution with the help of multiple cores.

Advantages of Multicore System

- These cores are usually integrated into single IC (integrated circuit) die, or onto multiple dies but in single chip package. Thus allowing higher Cache Coherency.
- These systems are energy efficient since they allow higher performance at lower energy. A challenge in this, however, is additional overhead of writing parallel code.
- It will have less traffic(cores integrated into single chip and will require less time).

Disadvantages of Multicore System

- Dual-core processor do not work at twice speed of single processor. They get only 60-80% more speed.
- Some Operating systems are still using single core processor.
- OS compiled for multi-core processor will run slightly slower on single-core processor.

What is MultiProcessor System?

Two or more processors or CPUs present in same computer, sharing system bus, memory and I/O is called MultiProcessing System. It allows parallel execution of different processors. These systems are reliable since failure of any single processor does not affect other processors. A quad-processor system can execute four processes at a time while an octa-processor can execute eight processes at a time. The memory and other resources may be shared or distributed among processes.

Advantages of MultiProcessor System

- Since more than one processor are working at the same time, throughput will get increased.
- More reliable since failure in one CPU does not affect other.
- It needs little complex configuration.

- Parallel processing (more than one process executing at same time) is achieved through MultiProcessing.

Disadvantages of MultiProcessor System

- It will have more traffic (distances between two will require longer time).
- Throughput may get reduced in shared resources system where one processor using some I/O then another processor has to wait for its turn.
- As more than processors are working at particular instant of time. So, coordination between these is very complex.

Difference Between MultiCore and MultiProcessor System

MultiCore	MultiProcessor
A single CPU or processor with two or more independent processing units called cores that are capable of reading and executing program instructions.	A system with two or more CPU's that allows simultaneous processing of programs.
It executes single program faster.	It executes multiple programs Faster.
Not as reliable as multiprocessor.	More reliable since failure in one CPU will not affect other.
It has less traffic.	It has more traffic.
It does not need to be configured.	It needs little complex configuration.
It's very cheaper (single CPU that does not require multiple CPU support system).	It is Expensive (Multiple separate CPU's that require system that supports multiple processors) as compared to MultiCore.

Conclusion

The primary goal of both multicore and multiprocessor systems is to enhance processing speed but they achieve this in different ways. Multicore systems are generally more costeffective because they integrate multiple cores within a single processor whereas multiprocessor systems require several physical processors driving up costs. If you're running a single program a multicore system will typically perform faster due to its efficient architecture. However if you need to run multiple programs simultaneously multiprocessor system shines by distributing the workload across several processors. In modern computing it's common to find computers equipped with multiple CPUs each containing several cores effectively combining the strengths of both architectures to meet diverse processing needs.

Challenges in programming for Multicore system

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Multicore System consists of two or more processors which have been attached to a single chip to enhance performance, reduce power consumption, and more efficient simultaneous processing of multiple tasks. Multicore system has been in recent trend where each core appears as a separate processor. Multicore system is capable of executing more than one threads parallelly whereas in Single core system only one thread can execute at a time.

Implementing Multicore system is more beneficial than implementing single core system by increasing number of transistors on single chip to enhance performance because increasing number of transistors on a single chip increases complexity of the system.

Challenges of multicore system :

Since multicore system consists of more than one processors, so the need is to keep all of them busy so that you can make better use of multiple computing cores. Scheduling algorithms must be designed to use multiple computing core to allow parallel computation. The challenge is also to modify the existing and new programs that are multithreaded to take advantage of multicore system.

In general five areas present challenges in programming for multicore systems :

1. **Dividing Activities :** The challenge is to examine the task properly to find areas that can be divided into separate, concurrent subtasks that can execute parallelly on individual processors to make complete use of multiple computing cores.

2. **Balance :** While dividing the task into sub-tasks, equality must be ensured such that every sub-task should perform almost equal amount of work. It should not be the case that one sub task has a lot of work to perform and other sub tasks have very less to do because in that case multicore system programming may not enhance performance compared to single core system.

3. **Data splitting :** Just as the task is divided into smaller sub-tasks, data accessed and manipulated by that task must also be divided to run on different cores so that data can be easily accessible by each sub-tasks.

4. **Data dependency :** Since various smaller sub-tasks run on different cores, it may be possible that one sub-task depends on the data from another sub tasks. So the

data needs to be examined properly so that the execution of whole task is synchronized.

5. **Testing and Debugging :** When different smaller sub-tasks are executing parallelly, so testing and debugging such concurrent tasks is more difficult than testing and debugging single threaded application.

Advantages and Disadvantages of Multicore Processors

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A multi-center CPU is a PC processor that has at least two segments. Each part of the chip executes guidelines as though it was a different PC. The genuine processors are as yet on one chip. On this chip, each center looks generally like the other. They are a few generally free centers that cooperate equally. A double center processor is a multi-center processor with two autonomous chips. A quad-center processor is a multi-center processor with four autonomous microchips. As you may have the option to tell from the prefix, the name of the processor depends on the quantity of the microchips on the chip. A multicore processor has numerous handling units (centers) on a solitary chip. Each center of the processor performs various errands. For instance, in the event that you are utilizing WhatsApp on the portable, at that point, one center is taking care of WhatsApp and the other centers of processor might be utilized for downloading an archive simultaneously. Outline of multi-center processor A multicore processor works like a human. Assume one human has one hand and another human has two hands. So one hand man can accomplish less work than two hands man. Additionally, the single-center processor can accomplish less work than a multi-center processor. The multicore processor likewise relies on the working framework utilized. Some working frameworks can't deal with multicore processors which need greater power. Assume you have a rapid processor then it will, at last, utilize greater power bringing about high PC battery utilization. On the off chance that you are playing high designs game, at that point it needs additionally handling force, and all the more preparing power implies greater power utilization so your PC battery will evaporate rapidly.

Multicore processors are utilized in the accompanying fields :

- Incredible illustrations arrangement
- PC supported plan (CAD)
- Sight and sound applications
- 3D gaming
- Video altering
- Information based workers
- Encoding

Advantages of multicore processors :

- Multicore processors can finish more work than single-center processors.
- Turns out incredible for multi-stringing applications.
- Can finish synchronous work as low recurrence.
- They can deal with more information than single-center processors.
- They can finish more work while burning through low energy when contrasted with the single-center processor.
- You can do complex works like filtering of the infection against infection and viewing a film simultaneously.
- As the two centers of processors are on a single chip so PC reserves exploits and information has not travel longer.
- PCB (printed circuit board) needs less space in the case of utilizing multi-core processors.
- Multicore processors are fault-tolerant to a great extent and are very reliable.
- Increased performance: Multicore processors can improve the performance of applications that are designed to run in parallel. This can result in faster processing times and better overall system performance.
- Reduced power consumption: Multicore processors can be more energy-efficient than single-core processors, as they can perform the same amount of work with less power.
- Improved multitasking: Multicore processors can improve multitasking performance by allowing multiple applications to run simultaneously on different cores.
- Enhanced reliability: Multicore processors can improve system reliability by providing redundant processing power. If one core fails, the other cores can continue to function, reducing the likelihood of system crashes.

Disadvantages of multicore processors :

- They are hard to oversee when contrasted with the single-center processor.
- They are expensive than a solitary center processor.
- Their speed isn't twice that of the typical processor.
- The presentation of the multicore processor relies on how the client utilizes the PC.
- They burn-through greater power.
- These processors become hot while accomplishing more work.
- On the off chance that some cycle needs direct/consecutive handling then the multicore processor needs to stand by longer.
- Complexity: Multicore processors can be more complex than single-core processors, as they require specialized software to take advantage of the multiple cores. This can make software development and system maintenance more difficult.
- Increased heat generation: Multicore processors can generate more heat than single-core processors, which can lead to higher power consumption and the need for more advanced cooling systems.
- Cost: Multicore processors can be more expensive than single-core processors, as they require more advanced technology to manufacture.
- Limited scalability: Multicore processors may not scale well beyond a certain number of cores, as the overhead associated with coordinating multiple cores can become a bottleneck.